Yihua Zhong / 钟益华

Ph.D. student in Intelligent Education · East China Normal University

Expected graduation: June 2028

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Research interests

Generative AI applications in education; agent-supported learning; personalized learning; learning analytics.

Education

East China Normal University — 2024.09–2028.06

Shanghai Institute of AI for Education · Ph.D. student in Intelligent Education (expected graduation: June 2028)

Zhejiang Normal University — 2021.09–2024.06

College of Education · Master's degree in Educational Technology

Zhejiang Provincial Key Laboratory of Intelligent Education Technology and Application

Jiangxi Normal University — 2017.09–2021.06

School of Software · Bachelor's degree in Software Engineering

Publications (9)

English renderings of Chinese publication and project titles are translations; original Chinese titles are retained below.

[1] Yihua Zhong, Jili Chen, Changqin Huang, Xizhe Wang, Yixuan Chen, Tao He (2026). An Explicit-Thinking LLM-Based Cognitive Apprenticeship Approach to Promote Students' Metacognition, Higher Order Thinking Skills, and Learning Performance. *IEEE Transactions on Learning Technologies, 19, 548–565*. (First author) https://doi.org/10.1109/TLT.2026.3696968

[2] 钟益华, 陈继力, 陈逸璇 (2026). 面向大语言模型系统能力培养的"三段式"实验教学设计——以ChatGLM-6B本地微调为例. 计算机教育, *(2), 23–28*. (First author) https://doi.org/10.16512/j.cnki.jsjjy.2026.02.041
English title: A three-stage laboratory teaching design for developing large language model systems competencies: A case study of local fine-tuning of ChatGLM-6B

[3] Changqin Huang, Yihua Zhong, Yongzhi Li, Xizhe Wang, Zhongmei Han, Di Zhang, Ming Liu (2025). Enhancing student reading performance through a personalized two-tier problem-based learning approach with generative artificial intelligence. *Humanities and Social Sciences Communications, 12, 645*. (Second author) https://doi.org/10.1057/s41599-025-04919-4

[4] 黄昌勤, 钟益华, 王希哲, 韩中美, 魏同权 (2025). 从单智能体到多智能体：大模型智能体支持下的激励型学习活动设计与实证研究. 华东师范大学学报（教育科学版）, *43(5), 44–56*. (Second author) https://doi.org/10.16382/j.cnki.1000-5560.2025.05.004

English title: From single-agent to multi-agent systems: Design and empirical study of motivational learning activities supported by large language model agents

[5] Xizhe Wang, Yihua Zhong, Changqin Huang, Xiaodi Huang (2024). ChatPRCS: A Personalized Support System for English Reading Comprehension Based on ChatGPT. *IEEE Transactions on Learning Technologies, 17, 1722–1736*. (Second author) https://doi.org/10.1109/tlt.2024.3405747

[6] Changqin Huang, Yihua Zhong, Xizhe Wang, Zhongmei Han, Tongquan Wei (2026). From Single-Agent System to Multi-Agent System: Design and Empirical Study on Motivational Learning Activities Supported by LLM-Based Agents. *Frontiers of Education in China, 21(2), 105–125*. (Second author) https://doi.org/10.3868/s110-021-026-0006-5

[7] Jili Chen, Yihua Zhong, Qionghao Huang, Changqin Huang, Fan Jiang, Xiaodi Huang, Xun Wang (2026). UCMIB-PNS: Balancing Sufficiency and Necessity With Probabilistic Causality and Cross-Modal Uncertainty in Multimodal Sentiment Analysis. *IEEE Transactions on Affective Computing, 17(1), 92–106*. https://doi.org/10.1109/taffc.2025.3606964

[8] Yixuan Chen, Changqin Huang, Zhongmei Han, Di Zhang, Yiheng Lou, Yihua Zhong (2026). Investigating student engagement with personalised generative AI feedback: Effects on creative problem-solving and feedback uptake. *Australasian Journal of Educational Technology, 42(4), 165–185*. https://doi.org/10.14742/ajet.11512

[9] Yixuan Chen, Xuemei Wu, Yihua Zhong, Changqin Huang (2026). Effects of Personalized Generative AI on Student Creativity: Evidence from Creative Performance, Self-efficacy, and Attention Allocation. *International Conference on Technology in Education (ICTE 2025). CCIS 2752, 376–388. Springer, 2026*. https://doi.org/10.1007/978-981-95-4499-8_31

Research projects and responsibilities

Participated in more than ten provincial/ministerial-level or higher research projects; selected projects as a core member are listed below.

1. Deep integration of artificial intelligence and education

Major Research Project in Philosophy and Social Sciences, Ministry of Education · Core member

Designing generative AI teaching models and conducting empirical research

2. Cognitive mechanisms and evolutionary patterns of knowledge construction through interpersonal interaction in teaching

Key Program, National Natural Science Foundation of China · Core member

Modeling learning interaction processes and analyzing cognitive mechanisms

3. Evidence-based validation methods for learners' cognitive and affective development across multiple settings

Subproject of the National Science and Technology Innovation Major Project on Next-Generation Artificial Intelligence · Core member

Analyzing cognitive and affective data and conducting evidence-based validation

4. Attributing learners' cognitive anomalies and providing adaptive guidance through collaborative LLM-based agents

General Program, National Natural Science Foundation of China · Core member

Designing multi-agent systems and investigating adaptive guidance

5. Attention-network-based cognitive load quantification and intervention in online education

Young Scientists Fund, National Natural Science Foundation of China · Core member

Modeling learning behavior and analyzing cognitive load

6. Assessment and intervention for teachers' online collaborative reflection using multimodal dynamic graph neural networks

Young Scientists Fund, National Natural Science Foundation of China · Core member

Modeling multimodal learning data and conducting assessment analyses

7. Innovation and practice of AI-enabled generative learning models in higher education

Major Humanities and Social Sciences Research Project for Universities in Zhejiang Province · Core member

Designing AI learning models and validating their application

8. Mechanisms and strategies for enhancing learners' higher-order thinking through generative AI

Key Project, Zhejiang Provincial Philosophy and Social Sciences Planning · Core member

Designing learning activities for higher-order thinking and analyzing underlying mechanisms

Teaching experience

Artificial Intelligence in Education — 2026.03–2026.07

East China Normal University · Teaching assistant

Assisted with teaching and practical instruction in generative AI, large language models, and intelligent education applications; guided students in designing and implementing AI applications for education.

Modern Educational Technology — 2023.09–2024.01

Zhejiang Normal University · Teaching assistant

Assisted with teaching and practical instruction in instructional design, digital educational resource development, and technology integration; guided students in creating instructional designs and digital learning resources.

Intellectual property

Method and system for personalized question generation and learning support based on large language models

Chinese invention patent application · CN202311484184.9 · Second inventor · Application accepted

Method, system, device, and medium for optimizing collaborative learning under load constraints

Chinese invention patent application · CN202210466712.7 · Second inventor · Application accepted

Deep-learning-based automatic question generation system V1.0

Computer software copyright · 2023SR0601805 · Second contributor · Registered

Academic service

Journal reviewer: Journal of Computer Assisted Learning; IEEE Transactions on Learning Technologies; Educational Technology Research and Development; Education and Information Technologies; Humanities & Social Sciences Communications; International Journal of Educational Technology in Higher Education; International Journal of Artificial Intelligence in Education; Computational Intelligence; Journal of Research in Reading.

More than 200 hours of peer review; Member, Chinese Association for Artificial Intelligence (CAAI).

Awards and honors

2021 — Jiangxi Normal University · Outstanding Graduate

2025、2026 — East China Normal University · First-Class Scholarship

2022、2023 — Zhejiang Normal University · First-Class Scholarship

2023 — Zhejiang Normal University · Chengcai Scholarship

2024 — Zhejiang Normal University · Yongquan Scholarship

Skills

Artificial Intelligence & Intelligent Education

Machine learning and deep learning; Large language models (LLMs); Generative artificial intelligence; Agent design and development.

Educational Technology & Instructional Design

Intelligent tutoring system design; AI-assisted instructional design; Personalized and adaptive learning; Intelligent learning environment design; Digital course and educational resource development.

Educational Data & Research Methods

Educational data analysis; Learning analytics; Experimental and quasi-experimental research; Survey research and statistical analysis; Mixed methods research; Data visualization.